

Connexions between traditional tests and mixed models

Brice Ozenne

October 21, 2025

This vignette connects well-known tests (t.test, ANCOVA, Pearson's correlation, Bartlett's test, ...) with the output of a linear mixed model. The connexion typically only holds in absence of missing data.

1 Illustrative datasets

We will consider two illustrative datasets:

- data from the abeta study lifestyle and psychosocial data between patients with newly diagnosed bipolar disorder (BD) and matched healthy controls (HC) at baseline: functioning assessment test (fast0), quality of life (qol0), perceived stress score (pss0), ... and at 1 year follow-up (pss1, fast1, qol1, for BD only).

```
data(abetaW, package = "LMMstar")
abetaW$missingreason <- NULL
head(abetaW)
```

	id	sex	age	group	episode	fast0	qol0	pss0	fast1	pss1	qol1	educationyears	alcohol
1	1	M	30	BD	0	1	88	9	0	NA	NA	13	0
2	2	F	55	BD	1	32	87	21	NA	NA	NA	15	0
3	3	M	51	BD	0	29	86	23	31	27	79	21	1
4	4	M	38	BD	0	1	96	7	6	6	101	21	1
5	5	M	21	BD	0	3	97	1	1	5	105	12	1
6	6	M	42	BD	1	22	70	17	40	18	68	13	0

This dataset shows great difference in heterogeneity between the two groups, e.g.:

```
library(LMMstar)
summarize(pss0 ~ group, data = abetaW, na.rm = TRUE)
```

	group	observed	missing	mean	sd	min	q1	median	q3	max
1	BD	86	1	13.2674	6.8435	1	7.25	13	17.75	29
2	HC	44	0	7.2727	5.0272	0	3.75	6	10.50	19

We will also use a balanced version of this dataset (equal group size):

```
abetaW.B <- do.call(rbind, by(abetaW, abetaW$group, function(iDF){
  iDF[which(!is.na(iDF$pss0))[1:44],]
}))
```

- data from the calcium dataset, a two-arm randomized clinical trial comparing bone mineral density between calcium supplement (C) and placebo (P). Visits were planned every 6 months, `bmd1` refers to the baseline measurement and `bmd2`, ..., `bmd5` refers to post-intervention measurements. `time.obs1`, ..., `time.obs5` refer to the time elapsed from baseline measurement in years.

```
data(calciumL, package = "LMMstar")
calciumL <- merge(by = "girl", calciumL,
                  transform(calciumL, baseline = bmd)[calciumL$visit==1,c("girl","baseline")])
calciumL <- calciumL[order(calciumL$girl,calciumL$visit),]
head(calciumL)
```

	girl	grp	dropout	dropvisit	visit	time.obs	bmd	baseline
1	101	C	0	NA	1	0.00000	815	815
3	101	C	0	NA	2	0.51472	875	815
5	101	C	0	NA	3	0.98015	911	815
2	101	C	0	NA	4	1.49760	952	815
4	101	C	0	NA	5	1.99589	970	815
10	102	P	0	NA	1	0.00000	813	813

The corresponding wide format is

```
data(calciumW, package = "LMMstar")
calciumW$dropout <- NULL
calciumW$dropvisit <- NULL
head(calciumW)
```

	girl	grp	bmd1	bmd2	bmd3	bmd4	bmd5	time.obs1	time.obs2	time.obs3	time.obs4	time.obs5
1	101	C	815	875	911	952	970	0	0.51472	0.98015	1.4976	1.9959
2	102	P	813	833	855	881	901	0	0.51472	0.95551	1.4730	1.9521
3	103	P	812	812	843	855	895	0	0.51198	0.95825	1.4757	1.9548
4	104	C	804	847	885	920	948	0	0.51198	0.97194	1.5086	2.1136
5	105	C	904	927	952	955	1002	0	0.57495	0.97741	1.4757	1.9548
6	106	P	831	855	890	908	933	0	0.53388	1.01300	1.5907	2.1684

We will use the placebo group as reference:

```
calciumW$grp <- relevel(calciumW$grp, "P")
calciumL$grp <- relevel(calciumL$grp, "P")
```

and the change from baseline in bone mineral density:

```
calciumW$change2 <- calciumW$bmd2 - calciumW$bmd1
calciumW$change3 <- calciumW$bmd3 - calciumW$bmd1
calciumW$change4 <- calciumW$bmd4 - calciumW$bmd1
calciumW$change5 <- calciumW$bmd5 - calciumW$bmd1
calciumL$change <- calciumL$bmd - calciumL$baseline
```

For illustrative purpose, we will restrict both dataset to subjects with complete data:

```
calciumW.NNA <- calciumW[rowSums(is.na(calciumW))==0,]
calciumL.NNA <- calciumL[calciumL$girl %in% calciumW.NNA$girl,]
```

as the aim is to show equivalence between statistical tests when there is no missing data.

2 Test on the mean

2.1 Welch two sample t-test

A two sample t-test:

```
with(abetaW, t.test(x = pss0[group=="BD"], y = pss0[group=="HC"]))
```

Welch Two Sample t-test

```
data: pss0[group == "BD"] and pss0[group == "HC"]
t = 5.67, df = 112, p-value = 1.1e-07
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 3.8988 8.0906
sample estimates:
mean of x mean of y
 13.2674    7.2727
```

is equivalent to a linear regression with a group-specific residual variance:

```
abetaW$group <- relevel(abetaW$group, "HC")
e.ttest <- lmm(pss0 ~ group, structure = IND(~group),
               data = abetaW, trace = FALSE)
model.tables(e.ttest, effects = "all")
```

	estimate	se	df	lower	upper	p.value
(Intercept)	7.2727	0.75788	43.009	5.7443	8.8011	2.9650e-12
groupBD	5.9947	1.05781	112.201	3.8988	8.0906	1.1399e-07
sigma	5.0272	0.54210	43.009	4.0447	6.2484	NA
k.BD	1.3613	0.18014	86.351	1.0464	1.7709	2.2090e-02

For comparison a linear model would estimate different standard errors, degrees of freedom, and p-values:

```
model.tables(lmm(pss0 ~ group, data = abetaW))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	7.2727	0.94857	128.03	5.3958	9.1496	3.8629e-12
groupBD	5.9947	1.16625	128.03	3.6871	8.3023	1.0000e-06

as it does not account for heteroschedasticity. This makes the 'heteroschedastic linear regression' `e.ttest` a natural extension of the t-test when it comes to account for covariates.

In the special case of two groups of equal size, the standard errors estimated accounting for heteroschedasticity:

```
model.tables(lmm(pss0 ~ group, structure = IND(~group),
                data = abetaW.B, trace = FALSE))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	11.8636	0.98648	43.009	9.8742	13.8530	2.4425e-15
groupHC	-4.5909	1.24399	80.661	-7.0662	-2.1156	4.0523e-04

or ignoring it:

```
model.tables(lmm(pss0 ~ group, data = abetaW.B))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	11.8636	0.87964	86.017	10.1150	13.6123	0.00000000
groupHC	-4.5909	1.24399	86.017	-7.0639	-2.1179	0.00039184

will be the same, leading to very similar p-values (degrees of freedom differ slightly).

2.2 Paired t-test

With complete data, a paired t-test:

```
t.test(calciumW.NNA$bmd2, calciumW.NNA$bmd1, paired = TRUE)
```

Paired t-test

```
data: calciumW.NNA$bmd2 and calciumW.NNA$bmd1
t = 13, df = 90, p-value <2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 20.229 27.529
sample estimates:
mean difference
 23.879
```

is equivalent to a LMM with an unstructured covariate pattern:

```
e.lmm2tt <- lmm(bmd ~ visit, repetition = ~visit|girl, structure = "UN",
               data = calciumL.NNA)
model.tables(e.lmm2tt)["visit2",,drop=FALSE]
```

	estimate	se	df	lower	upper	p.value
visit2	23.879	1.8371	89.968	20.229	27.529	0

2.3 Comparing change

2.3.1 Using a Welch two sample t-test

With complete data, a two sample t-test comparing the change from baseline:

```
ttc <- with(calciumW.NNA, t.test(x = change2[grp=="C"], y = change2[grp=="P"]))
ttc
```

Welch Two Sample t-test

```
data:  change2[grp == "C"] and change2[grp == "P"]
t = 2.03, df = 88.8, p-value = 0.046
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.14074 14.49659
sample estimates:
mean of x mean of y
 27.659    20.340
```

is equivalent to a LMM with a stratified unstructured covariate pattern:

```
e.lmm2tt2 <- lmm(bmd ~ visit*grp, repetition = ~visit|girl, structure = UN(~grp),
                data = calciumL.NNA)
model.tables(e.lmm2tt2)[c("visit2", "visit2:grpC"),,drop=FALSE]
```

	estimate	se	df	lower	upper	p.value
visit2	20.3404	2.5338	46.005	15.24013	25.441	2.6911e-10
visit2:grpC	7.3187	3.6124	88.734	0.14069	14.497	4.5767e-02

The estimate and standard error are exactly the same:

```
c(ttc$estimate["mean of x"] - ttc$estimate["mean of y"],
  se = ttc$stderr)
```

mean of x	se
7.3187	3.6124

The only (small) difference lies in the estimation of the degrees of freedom.

2.3.2 Using a linear regression

Using a linear model to compare change over time:

```
eLM.change <- lm(change2 ~ grp, data = calciumW.NNA)
summary(eLM.change)$coef
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	20.3404	2.5133	8.0931	2.7975e-12
grpC	7.3187	3.6144	2.0249	4.5878e-02

is equivalent to the following mixed model:

```
eLMM.change <- lmm(bmd ~ visit*grp,
                    repetition =~ visit|girl, structure = UN,
                    data = calciumL.NNA)
model.tables(eLMM.change)[c("visit2", "visit2:grpC"),]
```

	estimate	se	df	lower	upper	p.value
visit2	20.3404	2.5133	88.962	15.34654	25.334	2.8044e-12
visit2:grpC	7.3187	3.6144	88.962	0.13688	14.500	4.5880e-02

Here, since the linear regression assumes the same variance in both groups, we did not stratified the covariance pattern on group. The same equivalence would hold with a continuous exposure (say dose) instead of a binary exposure (here `grp`).

In presence of a covariate:

```
set.seed(1)
calciumW2.NNA <- cbind(calciumW.NNA,
                       age = round(runif(NROW(calciumW.NNA), min = 18, max = 60)))
calciumL2.NNA <- merge(calciumL.NNA, calciumW2.NNA[,c("girl", "age")], by = "girl")

eLMadj.change <- lm(change2 ~ grp + age, data = calciumW2.NNA)
summary(eLMadj.change)$coef
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.17495	6.68052	1.3734	0.173121
grpC	6.99548	3.57426	1.9572	0.053495
age	0.28771	0.15982	1.8002	0.075251

one should specify interaction with time in the mixed model to retrieve the same results:

```
eLMMadj.change <- lmm(bmd ~ visit*grp + visit*age,
                      repetition =~ visit|girl, structure = UN,
                      data = calciumL2.NNA)
model.tables(eLMMadj.change)[c("visit2", "visit2:grpC"),]
```

	estimate	se	df	lower	upper	p.value
visit2	9.1750	6.6805	87.966	-4.10126	22.451	0.173122
visit2:grpC	6.9955	3.5743	87.966	-0.10764	14.099	0.053497

2.4 Multiple Student's t-test

To adjust several t-tests for multiple testing, one can use the equivalence with `lmm`. This however require to specify the structure of the data (via the argument `repetition`), i.e., at which level replicates are independent so the software can deduce the appropriate number of independent observation across t-tests:

```
e.ttest2 <- lmm(change2 ~ grp, structure = IND(~grp),
               data = calciumW, repetition = ~1|girl, trace = FALSE)
e.ttest3 <- lmm(change3 ~ grp, structure = IND(~grp),
               data = calciumW, repetition = ~1|girl, trace = FALSE)
e.ttest4 <- lmm(change4 ~ grp, structure = IND(~grp),
               data = calciumW, repetition = ~1|girl, trace = FALSE)
e.ttest5 <- lmm(change5 ~ grp, structure = IND(~grp),
               data = calciumW, repetition = ~1|girl, trace = FALSE)
```

The `anova` method is then used to specify the parameter of interest and the results combined using `rbind`:

```
e.mttest <- rbind(anova(e.ttest2, effects = "grpC=0"),
                 anova(e.ttest3, effects = "grpC=0"),
                 anova(e.ttest4, effects = "grpC=0"),
                 anova(e.ttest5, effects = "grpC=0"))
model.tables(e.mttest, method = "single-step2")
```

	estimate	se	df	lower	upper	p.value
change2: grpC=0	6.7507	3.3549	103.014	-1.2191	14.721	0.112799
change3: grpC=0	13.8150	4.8336	95.812	2.3321	25.298	0.014660
change4: grpC=0	12.5190	5.8369	86.835	-1.3473	26.385	0.084529
change5: grpC=0	19.0155	6.4666	86.916	3.6533	34.378	0.011440

Note: the `single-step2` adjustment is similar to the `single-step` adjustment of the `multcomp` package, i.e., a max test adjustment. But instead of relying on the density of a multivariate Student's t-distribution, which requires equal degrees of freedom, it samples in a multivariate distribution with Student's t marginal possibly based on different degrees of freedom and a Gaussian copula. Being based on random sampling, results will slightly change everytime the code is run unless the initial state of the random number generator is set to a specific value before running the code:

```
set.seed(1)
model.tables(e.mttest, method = "single-step2")
```

	estimate	se	df	lower	upper	p.value
change2: grpC=0	6.7507	3.3549	103.014	-1.2151	14.717	0.113439
change3: grpC=0	13.8150	4.8336	95.812	2.3379	25.292	0.014590
change4: grpC=0	12.5190	5.8369	86.835	-1.3404	26.378	0.085339
change5: grpC=0	19.0155	6.4666	86.916	3.6609	34.370	0.011640

The `LMMstar.options` function can be used
to output the number of samples used:

and change it:

<code>LMMstar.options()\$n.sampleCopula</code>	<code>LMMstar.options(n.sampleCopula = 1e4)</code>
--	--

```
[1] 1e+05
```

This whole procedure can be streamlined using the long format and the `mlmm` function:

- the argument `by` indicates how to split the data. A separate model is fitted on each split.
- the argument `effects` indicates the test to be extracted for each model.
- the argument `name.short` is a cosmetic argument: should the name of each test be the covariate value or a combination of the covariate variable and the covariate value?

```
e.mttest2 <- mlmm(change ~ grp, structure = IND(~grp), repetition = ~visit|girl,  
                  data = calciumL[calciumL$visit!=1,], trace = FALSE,  
                  by = "visit", effects = "grpC=0", name.short = FALSE)  
set.seed(1)  
model.tables(e.mttest, method = "single-step2")
```

	estimate	se	df	lower	upper	p.value
change2: grpC=0	6.7507	3.3549	103.014	-1.2151	14.717	0.113439
change3: grpC=0	13.8150	4.8336	95.812	2.3379	25.292	0.014590
change4: grpC=0	12.5190	5.8369	86.835	-1.3404	26.378	0.085339
change5: grpC=0	19.0155	6.4666	86.916	3.6609	34.370	0.011640

The function `mlmm` can be used not only to emulate multiple t-tests but also for multiple linear regressions or linear mixed models. In the special case of multiple Welch two-sample test, a dedicated function `mt.test` offers a more user friendly interface:

```
set.seed(1)  
mt.test(change2 + change3 + change4 + change5 ~ grp, data = calciumW)
```

Argument 'data' contains 59 missing values.

	estimate	se	df	lower	upper	p.value
change2	6.7507	3.3549	103.014	-1.2151	14.717	0.113439
change3	13.8150	4.8336	95.812	2.3379	25.292	0.014590
change4	12.5190	5.8369	86.835	-1.3404	26.378	0.085339
change5	19.0155	6.4666	86.916	3.6609	34.370	0.011640

2.5 ANCOVA

Instead of comparing the final value or the change between groups using a Welch two sample t-test, the ANCOVA is often referred to as the superior approach to assess a treatment effect (Vickers and Altman, 2001). It regresses the group variable and the baseline value against the change:

```
model.tables(lmm(change2 ~ bmd1 + grp, data = calciumW.NNA))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	-25.742684	25.757918	88.018	-76.930991	25.44562	0.320337
bmd1	0.052948	0.029457	88.018	-0.005592	0.11149	0.075693
grpC	6.741021	3.584377	88.018	-0.382155	13.86420	0.063324

or the final value:

```
model.tables(lmm(bmd2 ~ bmd1 + grp, data = calciumW.NNA))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	-25.7427	25.757918	88.018	-76.93099	25.4456	0.320337
bmd1	1.0529	0.029457	88.018	0.99441	1.1115	0.000000
grpC	6.7410	3.584377	88.018	-0.38215	13.8642	0.063324

both leading to equivalent result. The corresponding mixed model constrains the both group to take the same baseline value. This can be specified by introducing a new covariate that only differ between groups after baseline:

```
calciumL.NNA$trt <- ifelse(calciumL.NNA$visit==1,"P",as.character(calciumL.NNA$grp))
calciumL.NNA$trt <- factor(calciumL.NNA$trt, levels = c("P","C"))
ftable(grp = calciumL.NNA$grp, trt = calciumL.NNA$trt, visit = calciumL.NNA$visit)
```

	visit	1	2	3	4	5
grp trt						
P P		47	47	47	47	47
C		0	0	0	0	0
C P		44	0	0	0	0
C		0	44	44	44	44

We then retrieve the same estimate and similar (but not identical) standard errors and p-values with the following mixed model:

```
e.lmmANCOVA <- lmm(bmd ~ visit*trt, repetition = ~visit|girl, structure = UN,
                    data = calciumL.NNA)
model.tables(e.lmmANCOVA)["visit2:trtC",,drop=FALSE]
```

Constant values in the design matrix for the mean structure.

Coefficient "trtC" relative to interaction "visit:trt" has been removed.

	estimate	se	df	lower	upper	p.value
visit2:trtC	6.741	3.5642	88.853	-0.3411	13.823	0.061839

To avoid the message about the design matrix, one should 'manually' define the interaction terms:

```
calciumL.NNA$visit.trt <- ifelse(calciumL.NNA$trt == "C", calciumL.NNA$visit, "baseline")
calciumL.NNA$visit.trt <- factor(calciumL.NNA$visit.trt, levels = c("baseline",2:5))
ftable(grp = calciumL.NNA$grp, visit.trt = calciumL.NNA$visit.trt, visit = calciumL.NNA$visit)
```

		visit	1	2	3	4	5
grp	visit.trt						
P	baseline		47	47	47	47	47
	2		0	0	0	0	0
	3		0	0	0	0	0
	4		0	0	0	0	0
	5		0	0	0	0	0
C	baseline		44	0	0	0	0
	2		0	44	0	0	0
	3		0	0	44	0	0
	4		0	0	0	44	0
	5		0	0	0	0	44

```
e.lmmANCOVA2 <- lmm(bmd ~ visit + visit.trt, repetition = ~visit|girl, structure = UN,
                     data = calciumL.NNA)
model.tables(e.lmmANCOVA2) ["visit.trt2",,drop=FALSE]
```

	estimate	se	df	lower	upper	p.value
visit.trt2	6.741	3.5642	88.853	-0.3411	13.823	0.061839

As before, in presence of a covariate:

```
summary(lm(bmd2 ~ bmd1 + grp + age, data = calciumW2.NNA))$coef
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-37.6452	26.215165	-1.4360	1.5459e-01
bmd1	1.0536	0.029064	36.2524	2.7566e-54
grpC	6.4062	3.540822	1.8093	7.3865e-02
age	0.2914	0.157689	1.8479	6.8008e-02

one should add the covariate along with time interactions to retrieve the same estimate and similar standard error/p-value/confidence intervals with a linear mixed model:

```
calciumL2.NNA$visit.trt <- ifelse(calciumL2.NNA$grp == "C", calciumL.NNA$visit, "1")
e.lmmANCOVAadj <- lmm(bmd ~ visit + visit.trt + visit*age, repetition = ~visit|girl,
                      structure = UN, data = calciumL2.NNA)
model.tables(e.lmmANCOVAadj) ["visit.trt2",,drop=FALSE]
```

	estimate	se	df	lower	upper	p.value
visit.trt2	6.4062	3.5206	87.855	-0.59046	13.403	0.072223

A natural extension of the ANCOVA would be to relax the assumption of common residual variance between the two treatment groups:

```
model.tables(lmm(change2 ~ bmd1 + grp, data = calciumW.NNA, structure = IND(~grp)))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	-25.833272	25.805339	83.926	-77.1506784	25.48413	0.319665
bmd1	0.053052	0.029513	84.179	-0.0056359	0.11174	0.075828
grpC	6.739886	3.585265	87.584	-0.3855470	13.86532	0.063448

However the 'straightforward' connexion with mixed model seems lost:

```
e.lmmHANCova <- lmm(bmd ~ visit + visit.trt, repetition = ~visit|girl, structure = UN(~grp),
  data = calciumL.NNA)
model.tables(e.lmmHANCova)["visit.trt2",,drop=FALSE]
```

	estimate	se	df	lower	upper	p.value
visit.trt2	6.7516	3.5654	88.326	-0.33341	13.837	0.061542

3 Test on the correlation

3.1 Person's correlation

One can retrieve Pearson's correlation:

```
cor.test(calciumW.NNA$bmd1,calciumW.NNA$bmd5)
```

Pearson's product-moment correlation

```
data: calciumW.NNA$bmd1 and calciumW.NNA$bmd5
t = 18.3, df = 89, p-value <2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.83615 0.92551
sample estimates:
      cor
0.88901
```

using a linear mixed model moving to the long format and using an unstructured mean and covariance pattern over time:

```
eCor.lmm <- lmm(bmd ~ visit, repetition = ~visit|girl,
               structure = UN, data = calciumL.NNA)
model.tables(eCor.lmm, effects = "correlation")["rho(1,5)",]
```

	estimate	se	df	lower	upper	p.value
rho(1,5)	0.88901	0.0221	96.839	0.83607	0.92555	0

P-value and confidence interval will differ (only slightly in large samples) because `cor.test` uses an exact¹ formula for the variance after `atanh` transformation while the linear mixed model rely on the observed information matrix. In this example the observed information (default option) is more in line with `cor.test` than the expected information:

```
model.tables(eCor.lmm, type.information = "expected", effects = "correlation")["rho(1,5)",]
```

	estimate	se	df	lower	upper	p.value
rho(1,5)	0.88901	0.021914	17285033	0.83738	0.92492	0

Of note the confidence intervals and p-value of `cor.test` are not computed in a consistent way:

```
set.seed(7303)
X <- rnorm(10)
Y <- rnorm(10)
cor.test(X,Y)
```

¹assuming jointly normally distributed outcomes

Pearson's product-moment correlation

```
data: X and Y
t = 2.29, df = 8, p-value = 0.051
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.00016154 0.90179629
sample estimates:
      cor
0.62972
```

Here the confidence intervals do not overlap 0, i.e., suggest to reject the null hypothesis while the p-value is greater than 0.05, i.e., does not suggest to reject the null hypothesis. The corresponding mixed model estimate:

```
dfXY <- rbind(data.frame(value = X, variable = "x", id = 1:10),
              data.frame(value = Y, variable = "y", id = 1:10))
e.lmmXY <- lmm(value ~ variable, repetition = ~variable|id,
               structure = UN, data = dfXY)
model.tables(e.lmmXY, effects = "correlation")
```

```
      estimate      se    df    lower  upper p.value
rho(x,y)  0.62972 0.20115 7.0024 -0.047159 0.91027 0.061602
```

is the same but the confidence intervals and p-value differ more substantially (due to small sample approximations). They however are consistent with respect to whether to reject the null hypothesis.

3.2 Comparing Person's correlations

To compare the Pearson's correlation between two groups, one can use Fisher's z test:

```
rho.C <- with(calciumW.NNA, cor(bmd1[grp=="C"],bmd5[grp=="C"]))
rho.P <- with(calciumW.NNA, cor(bmd1[grp=="P"],bmd5[grp=="P"]))
nobs.C <- sum(calciumW$grp=="C")
nobs.P <- sum(calciumW$grp=="P")
stat.fisher <- (atanh(rho.C) - atanh(rho.P))/sqrt(1/(nobs.C-3)+1/(nobs.P-3))
2*(1-pnorm(abs(stat.fisher)))
```

```
[1] 0.15261
```

and the confidence intervals suggested by [Zou \(2007\)](#):

```
zou.C <- tanh(atanh(rho.C) + qnorm(c(0.025,0.975))/sqrt(nobs.C-3))
zou.P <- tanh(atanh(rho.P) + qnorm(c(0.025,0.975))/sqrt(nobs.P-3))

(rho.C - rho.P) - sqrt( (rho.C-zou.C[1])^2 + (rho.P-zou.P[2])^2 )
(rho.C - rho.P) + sqrt( (rho.C-zou.C[2])^2 + (rho.P-zou.P[1])^2 )
```

```
[1] -0.15309
```

```
[1] 0.021034
```

which is implemented in the package cocor:

```
library(cocor)
cocor.indep.groups(r1.jk = rho.C, n1 = nobs.C, r2.hm = rho.P, n2 = nobs.P)
```

Results of a comparison of two correlations based on independent groups

Comparison between r1.jk = 0.8597 and r2.hm = 0.917

Difference: r1.jk - r2.hm = -0.0574

Group sizes: n1 = 55, n2 = 57

Null hypothesis: r1.jk is equal to r2.hm

Alternative hypothesis: r1.jk is not equal to r2.hm (two-sided)

Alpha: 0.05

fisher1925: Fisher's z (1925)

z = -1.4304, p-value = 0.1526

Null hypothesis retained

zou2007: Zou's (2007) confidence interval

95% confidence interval for r1.jk - r2.hm: -0.1531 0.0210

Null hypothesis retained (Interval includes 0)

We can retrieve the same estimated difference and similar but not identical CIs/p-values using a linear mixed model with a covariance pattern stratified on group:

```
eCor2.lmm <- lmm(bmd ~ visit*grp, repetition = ~visit|girl,
                 structure = UN(~grp), data = calciumL.NNA)
model.tables(eCor2.lmm, effects = "correlation")[c("rho(1,5):C", "rho(1,5):P"),]
```

```
      estimate      se    df  lower  upper  p.value
rho(1,5):C  0.85965 0.039801 42.111 0.75492 0.92163 1.2128e-10
rho(1,5):P  0.91701 0.023456 53.835 0.85496 0.95319 7.3275e-15
```

and use a Wald test to compare the correlation coefficients:

```
set.seed(1)
summary(anova(eCor2.lmm, effects = "rho(1,5):C - rho(1,5):P = 0"), digits = 4)
```

Wald F-test

```
      statistic      df p.value
all      1.7165 (1,93.6)  0.193
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
df: Satterthwaite approximation w.r.t. model-based se.
```

Emulated Wald test (resampling parameter distribution)

```
      estimate      se  df  lower upper p.value
rho(1,5):C - rho(1,5):P = 0 -0.0574 0.0495 <NA> -0.1661  0.03  0.197
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
se: based on the observed information (model-based).
Back-transformation: rho parameters with atanh (1e+05 samples).
```

The 'Wald F-test' is the usual Wald test defined by the squared difference between the two correlation coefficients divided by the squared standard error of this difference. This ratio follows, under the null hypothesis, an F-distribution which is used to obtain a p-value. The 'Emulated Wald test' attempts to provide a confidence interval for the difference compatible with the p-value. As mentioned in the literature (Zou, 2007), a 'naive' back-transformation of the difference would not provide confidence intervals with good frequentist properties (intuitively $\tanh(\operatorname{atanh}(y) - \operatorname{atanh}(x)) \neq y - x$). Instead samples are drawn from a bivariate Student's t distribution centered around 0 and with variance-covariance matrix the inverse of the observed information on the `atanh` scale.

- **p.value**: relative frequency of a difference in simulated correlations more extreme than observed. It should be close to the p-value of the Wald F-test'.
- **se**: standard deviation of the simulated difference in correlation on the original scale
- **lower, upper**: quantiles of the simulated difference in correlation on the original scale after centering the simulated values on the `atanh` scale around the estimated correlation.

The `partialCor` method provides a more straightforward syntax to do the later test is:

```
set.seed(1)
partialCor(bmd1 + bmd5 ~ 1, data = calciumW.NNA, by = "grp", effects = "Dunnett")
```

```
      estimate      se df  lower upper p.value
C - P  -0.0574 0.0491 NA  -0.165 0.029   0.195
```

The methodology is the same, except that the underlying mixed model is based on two timepoints (1 and 5) instead of all timepoints (1,2,3,4,5).

It is also possible to not use any transformation:

```
testRho <- anova(eCor2.lmm, effects = "rho(1,5):C - rho(1,5):P = 0", transform.rho = "none")
summary(testRho, print = TRUE)
```

Wald F-test

```
      statistic      df p.value
all      1.542 (1,3.7)   0.288
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
df: Satterthwaite approximation w.r.t. model-based se.
```

Hypothesis-specific Wald test

```
      estimate      se df  lower upper p.value
rho(1,5):C - rho(1,5):P = 0  -0.057 0.046 3.7 -0.19 0.076   0.288
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
df: Satterthwaite approximation w.r.t. model-based se.
se: based on the observed information (model-based).
```

but this is expected to have worse small sample properties compared to using a transformation. In this example the estimated p-value is also further away from the Fisher's z test. Here the 'Hypothesis-specific Wald test' uses a Student's t-distribution to model the distribution of the ratio between the estimate and the standard error. This is exactly the square root (up to a sign) of the Wald F-test test statistic, leading to exactly the same p-value and compatible confidence intervals.

3.3 Correlation between changes

In some studies, one is interested in studying the relation between two evolutions. Say the change from baseline in quality of life vs. functioning assessment test:

```
abetaW$dqol <- abetaW$qol1 - abetaW$qol0
abetaW$dfast <- abetaW$fast1 - abetaW$fast0
abetaW.NNA <- abetaW[!is.na(abetaW$dqol) & !is.na(abetaW$dfast),]
```

One can evaluate their correlation:

```
cor.test(abetaW.NNA$dqol, abetaW.NNA$dfast)
```

Pearson's product-moment correlation

```
data: abetaW.NNA$dqol and abetaW.NNA$dfast
t = -4.27, df = 110, p-value = 4.2e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.52570 -0.20575
sample estimates:
      cor
-0.37692
```

or estimate the regression coefficient of one change against the other:

```
model.tables(lmm(dqol ~ dfast, data = abetaW.NNA))
```

	estimate	se	df	lower	upper	p.value
(Intercept)	1.34601	0.85087	110.02	-0.34022	3.03224	1.1654e-01
dfast	-0.49231	0.11535	110.02	-0.72091	-0.26371	4.1977e-05

To retrieve the same results using a linear mixed model, one should move the dataset to the very long format, where each type of measurement is treated as a separate outcome:

```
abetaL.NNA <- reshape(abetaW.NNA[,c("id","qol0","qol1","fast0","fast1")], direction = "long",
                      idvar = "id", varying = 2:5,
                      timevar = "type", times = c("qol0","qol1","fast0","fast1"), v.names = c(
                        "value"))
abetaL.NNA <- abetaL.NNA[order(abetaL.NNA$id),]
rownames(abetaL.NNA) <- NULL
head(abetaL.NNA)
```

	id	type	value
1	3	qol0	86
2	3	qol1	79
3	3	fast0	29
4	3	fast1	31
5	4	qol0	96
6	4	qol1	101

One can then jointly model the association between all type of measurement using an unstructured residual variance-covariance matrix:

```
e.lmm4 <- lmm(value ~ type,
              repetition = ~type|id, structure = "UN",
              data = abetaL.NNA)
sigma.lmm4 <- sigma(e.lmm4)
sigma.lmm4
```

```
      fast0  fast1  qol0  qol1
fast0 132.471  95.090 -97.958 -72.709
fast1  95.090 102.301 -75.656 -72.360
qol0  -97.958 -75.656 143.759  91.321
qol1  -72.709 -72.360  91.321 114.957
```

Deduce the residual covariance matrix for the change:

```
Mcon <- cbind(c(-1,1,0,0),c(0,0,-1,1))
sigmeChange.lmm4 <- t(Mcon) %*% sigma.lmm4 %*% Mcon
dimnames(sigmeChange.lmm4) <- replicate(2,c("dfast","dqol"), simplify = FALSE)
sigmeChange.lmm4
```

```
      dfast  dqol
dfast 44.592 -21.953
dqol  -21.953 76.075
```

and retrieve the correlation and regression coefficients:

```
cov2cor(sigmeChange.lmm4)[1,2]
sigmeChange.lmm4[1,2]/sigmeChange.lmm4[1,1]
```

```
[1] -0.37692
[1] -0.49231
```

The uncertainty can be quantified using a delta method:

```
estimate(e.lmm4, function(p){
  Sigma.change <- t(Mcon) %*% sigma(e.lmm4, p = p) %*% Mcon
  c(cor = cov2cor(Sigma.change)[1,2],
    beta = Sigma.change[1,2]/Sigma.change[1,1])
})
```

```
      estimate      se    df  lower  upper  p.value
cor  -0.37692 0.081429 12.075 -0.55421 -0.19962 0.00057192
beta -0.49231 0.114833 12.561 -0.74127 -0.24334 0.00095359
```

The standard error for the regression coefficient is close to the linear model one but the degrees of freedom seem grossly underestimated. One can set the argument `df` to `FALSE` when calling `estimate` to use a Gaussian instead of a Student's t distribution.

4 Test on the variance

4.1 Comparing variances

We can emulate a F-test comparing the variance between two populations:

```
var.test(x = calciumW.NNA[calciumW.NNA$grp=="C","bmd1"],
         y = calciumW.NNA[calciumW.NNA$grp=="P","bmd1"])
```

F test to compare two variances

```
data: calciumW.NNA[calciumW.NNA$grp == "C", "bmd1"] and calciumW.NNA[calciumW.NNA$grp == "P", "bmd1"]
F = 0.666, num df = 43, denom df = 46, p-value = 0.18
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.36801 1.21107
sample estimates:
ratio of variances
 0.66559
```

using an heteroschedastic linear regression with a parameter for the residual standard deviation in the reference group (σ) and a parameter for the ratio in standard deviation between the two groups (k):

```
eVar2.lmm <- lmm(bmd1 ~ grp, structure = IND(~grp), data = calciumW.NNA)
coef(eVar2.lmm, effects = "variance")
```

```
sigma      k.C
66.87928   0.81584
```

This leads to the following modeled group-sepecific residual standard deviations:

```
coef(eVar2.lmm, effects = "variance", transform.k = "sd")
```

```
sigma.P sigma.C
66.879   54.563
```

Testing whether the k parameter is 1, i.e. its log is 0:

```
summary(anova(eVar2.lmm, effects = "variance"))
```

Hypothesis-specific Wald test

```
              estimate      se    df lower upper p.value
variance: k.C=1    0.816 0.122 88.6  0.606 1.099    0.178
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
df: Satterthwaite approximation w.r.t. model-based se.
se: based on the observed information (model-based).
Back-transformation: k parameters with exp.
```

leads to a similar p-value compared to `var.test`. The estimate differs as `anova` returns the ratio of the residual standard deviations instead of the ratio of the residual variances. The latter can be obtained using:

```
summary(anova(eVar2.lmm, effects = "variance", transform.k = "logsquare"))
```

Hypothesis-specific Wald test

```

              estimate se    df lower upper p.value
variance: k.C=1    0.666 0.2 88.6 0.367 1.208    0.178
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
df: Satterthwaite approximation w.r.t. model-based se.
se: based on the observed information (model-based).
Back-transformation: k parameters with exp.
```

closely matching the output of `var.test`. This test is the special case of the Bartlett test:

```
bartlett.test(bmd1 ~ grp, data = calciumW.NNA)
```

Bartlett test of homogeneity of variances

```

data:  bmd1 by grp
Bartlett's K-squared = 1.8, df = 1, p-value = 0.18
```

which generalizes to more than two variances:

```

bartlett.test(age ~ sex.group,
              data = transform(abetaW.NNA, sex.group = paste0(sex,group)))
```

Bartlett test of homogeneity of variances

```

data:  age by sex.group
Bartlett's K-squared = 1.68, df = 3, p-value = 0.64
```

An F-test from the corresponding heteroscedastic linear regression leads to the same results:

```

eVar4.lmm <- lmm(age ~ sex.group, structure = IND(~sex.group),
                 data = transform(abetaW.NNA, sex.group = paste0(sex,group)))
summary(anova(eVar4.lmm, effects = "variance"))
```

Wald F-test

```

              statistic      df p.value
variance: sex.group    0.566 (3,53.1)    0.64
-----
:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
```

```
df: Satterthwaite approximation w.r.t. model-based se.
Multiple testing adjustment: joint test.
```

Hypothesis-specific Wald tests

	estimate	se	df	lower	upper	p.value
variance: k.FHC=1	1.143	0.241	34.3	0.681	1.919	0.874
k.MBD=1	0.92	0.157	68.5	0.606	1.398	0.935
k.MHC=1	1.131	0.22	47.2	0.701	1.822	0.876

```
: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.
df: Satterthwaite approximation w.r.t. model-based se.
se: based on the observed information (model-based).
Multiple testing adjustment: max test (1e+05 samples).
Back-transformation: k parameters with exp.
```

Note that the test statistic of `anova` multiplied by its (numerator) degree of freedom 0.566×3 leads to the test statistic of `bartlett.test`.

⚠ when considering variance of time instead of variance between groups the equivalence is typically lost as `lmm` can account for within-subject correlation (argument `structure` set to `UN`) while `bartlett.test` cannot.

References

- Vickers, A. J. and Altman, D. G. (2001). Analysing controlled trials with baseline and follow up measurements. *Bmj*, 323(7321):1123–1124.
- Zou, G. Y. (2007). Toward using confidence intervals to compare correlations. *Psychological methods*, 12(4):399.